

99.500 Applied Mathematics for Engineering (Spring AY2026)

Modular Credit: 12 & Workload: 3-0-9 or 3-2-7

Instructor: Dr. Hansong Xue (hansong_xue@sutd.edu.sg, Office: 1.302.24)

Regular Class Time: Wednesday, 2:00pm – 5:00pm, Think Tank 22 (2.311)

Consultation Hours: Tuesday, 3:00pm – 4:00pm or by appointment.

Learning Objectives

At the end of the term, students will be able to:

1. Understand the fundamentals of complex analysis and apply the tools to evaluate integrals and solve complex equations appearing in engineering problems.
2. Understand Laplace and Fourier transforms and apply them to solve differential equations.
3. Understand the fundamentals of ordinary and partial differential equations, be familiar with several important differential equations encountered in engineering problems and know the basic techniques to solve them.

In addition, students will also have the opportunity to enhance the following soft skills, which are increasingly valuable for postgraduate students, through participation in class activities:

1. Communication skills, including oral and written presentations, as well as understanding teamwork dynamics.
2. Collaboration skills, particularly in group settings, including conflict resolution.
3. Critical and creative thinking when solving open-ended problems.

Target Audience and Pre-requisites

Postgraduate and undergraduate students interested in mathematical modelling and applied mathematical tools useful for engineers. A background in complex analysis, solving differential equations, and computer programming will be helpful, though not required.

Assessment

Participation (10%) (Surveys, individual meetings, bonus questions/tasks, etc.)

Homework assignments (10%)

Mid-term group project (30%) (a comprehensive self-constructed written exam)

Computer-based lab work (20% in Week 13)

Final exam (30%)

Content Coverage and Tentative Class Schedule

Week 1: Course introduction, pre-module survey, syllabus and policies

Week 2: Modeling with deriving differential equations and non-dimensionalization (HW1)

Week 3: Ordinary differential equations (ODEs) and analytical solution of ODEs, mid-term project instructions

Week 4: No class due to CNY holiday

Week 5: Bessel ODEs (HW2)

Week 6: Numerical solution of ODEs (Matlab lab session)

----- Week 7: Recess Week (individual meetings + mid-term 1st draft submission) -----

Week 8: Mid-term project presentation, discussion and feedback

Week 9: Partial differential equations (PDEs) with separation of variables (SOV) (HW3)

Week 10: PDEs Laplace Transform and final submission of mid-term project report (HW4)

Week 11: PDEs Fourier transform and complex analysis (HW5)

Week 12: Numerical solution of PDEs (Matlab lab session) and discussion of exam questions

Week 13: Computer test, review of semester and exam instructions

Week 14: Final Exam (29 April, Wednesday 2:00 PM – 4:00 PM) and individual meetings

Recommended Text and Readings

- Advanced Engineering Mathematics, Erwin Kreyszig, 10th Edition, John Wiley & Sons, Inc.
- Applied Mathematics for Engineers, Ramin S. Esfandiari, 6th Edition, Atlantis Publishing Company
- Applied Numerical Methods Using Matlab®, Won Y. Yang et al., 2nd Edition, John Wiley & Sons, Inc.
- Applied Mathematics and Modeling For Chemical Engineers, Richard G. Rice & Duong D. Do, 3rd Edition, John Wiley & Sons, Inc.
- Complex Variables and Applications, James Ward Brown & Ruel V. Churchill, 7th Edition, McGraw-Hill

Other Policies

1. The group work in this course is designed to facilitate collaborative learning and learning from peers; it does NOT mean to “share/decrease the workload.”
2. Groups have to be formed by Week 2 class. Changes are not allowed in principle, except in cases where a student quits the course. The instructor will reassign groups if necessary.
3. Homework should be done in groups of 2-3 students. Late submissions without a valid reason will result in a reduction of marks. The deadline for submission is one week after the release of the homework.
4. For those who are not familiar with MATLAB, the lab sessions will be a good opportunity to learn.
5. The mid-term exam will be administered in a “self-constructed” mode: take-home, open-resource, and group work. More details can be found in the “Guidelines for Mid-term Exam.”
6. Two required individual meetings (in Weeks 7 and 14) will be scheduled for instructor-student interaction. The instructor values this opportunity to learn more about students’ learning needs, expectations, and opinions on teaching to provide a better learning experience.